chcp 65001

SET PGCLIENTENCODING=UTF8

SET PAGER=

psql -U postgres -d beer_import_db -f queries.sql

psql -U postgres -d beer_import_db -f indexes.sql

```
E:\BD2026>chcp 65001
Active code page: 65001

E:\BD2026>SET PGCLIENTENCODING=UTF8

E:\BD2026>SET PAGER=

E:\BD2026>psql -U postgres -d beer_import_db -f queries.sql
SET
SET
---1---
                                                
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---5---
                                                                    QUERY PLAN
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```
Sort  (cost=18.67..18.68 rows=6 width=649) (actual time=0.160..0.162 rows=3.00 loops=1)
  Sort Key: (sum(sd.day_sales_rub)) DESC
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=4
  CTE sales_detail
    ->  HashAggregate  (cost=5.97..6.06 rows=6 width=72) (actual time=0.064..0.067 rows=6.00 loops=1)
          Group Key: dc.venue_id, td.transfer_date
          Batches: 1  Memory Usage: 32kB
          Buffers: shared hit=3
          ->  Hash Join  (cost=4.10..5.89 rows=6 width=19) (actual time=0.049..0.054 rows=6.00 loops=1)
                Hash Cond: (dc.id = td.contract_id)
                Buffers: shared hit=3
                ->  Seq Scan on delivery_contract dc  (cost=0.00..1.42 rows=42 width=8) (actual time=0.005..0.006 rows=42.00 loops=1)
                      Buffers: shared hit=1
                ->  Hash  (cost=4.02..4.02 rows=6 width=19) (actual time=0.039..0.040 rows=6.00 loops=1)
                      Buckets: 1024  Batches: 1  Memory Usage: 9kB
                      Buffers: shared hit=2
                      ->  Hash Join  (cost=2.04..4.02 rows=6 width=19) (actual time=0.028..0.036 rows=6.00 loops=1)
                            Hash Cond: (tdi.document_id = td.id)
                            Buffers: shared hit=2
                            ->  Seq Scan on transfer_document_item tdi  (cost=0.00..1.77 rows=77 width=15) (actual time=0.004..0.006 rows=77.00 loops=1)
                                  Buffers: shared hit=1
                            ->  Hash  (cost=1.96..1.96 rows=6 width=12) (actual time=0.018..0.018 rows=6.00 loops=1)
                                  Buckets: 1024  Batches: 1  Memory Usage: 9kB
                                  Buffers: shared hit=1
                                  ->  Seq Scan on transfer_document td  (cost=0.00..1.96 rows=6 width=12) (actual time=0.010..0.015 rows=6.00 loops=1)
                                        Filter: (transfer_date >= '2026-01-01'::date)
                                        Rows Removed by Filter: 71
                                        Buffers: shared hit=1
  ->  GroupAggregate  (cost=12.36..12.54 rows=6 width=649) (actual time=0.149..0.155 rows=3.00 loops=1)
        Group Key: v.id, gf.has_long_gap
        Buffers: shared hit=4
        ->  Sort  (cost=12.36..12.37 rows=6 width=585) (actual time=0.145..0.147 rows=6.00 loops=1)
              Sort Key: v.id, gf.has_long_gap
              Sort Method: quicksort  Memory: 25kB
              Buffers: shared hit=4
              ->  Hash Left Join  (cost=0.83..12.28 rows=6 width=585) (actual time=0.134..0.142 rows=6.00 loops=1)
                    Hash Cond: (v.id = gf.venue_id)
                    Buffers: shared hit=4
                    ->  Hash Join  (cost=0.20..11.63 rows=6 width=584) (actual time=0.092..0.098 rows=6.00 loops=1)
                          Hash Cond: (v.id = sd.venue_id)
                          Buffers: shared hit=4
                          ->  Seq Scan on venue v  (cost=0.00..11.00 rows=100 width=520) (actual time=0.013..0.015 rows=33.00 loops=1)
                                Buffers: shared hit=1
                          ->  Hash  (cost=0.12..0.12 rows=6 width=68) (actual time=0.073..0.073 rows=6.00 loops=1)
                                Buckets: 1024  Batches: 1  Memory Usage: 9kB
                                Buffers: shared hit=3
                                ->  CTE Scan on sales_detail sd  (cost=0.00..0.12 rows=6 width=68) (actual time=0.066..0.071 rows=6.00 loops=1)
                                      Storage: Memory  Maximum Storage: 17kB
                                      Buffers: shared hit=3
                    ->  Hash  (cost=0.56..0.56 rows=6 width=5) (actual time=0.033..0.033 rows=3.00 loops=1)
                          Buckets: 1024  Batches: 1  Memory Usage: 9kB
                          ->  Subquery Scan on gf  (cost=0.20..0.56 rows=6 width=5) (actual time=0.024..0.029 rows=3.00 loops=1)
                                ->  GroupAggregate  (cost=0.20..0.50 rows=6 width=5) (actual time=0.024..0.028 rows=3.00 loops=1)
                                      Group Key: sales_detail.venue_id
                                      ->  WindowAgg  (cost=0.20..0.33 rows=6 width=12) (actual time=0.019..0.024 rows=6.00 loops=1)
                                            Window: w1 AS (PARTITION BY sales_detail.venue_id ORDER BY sales_detail.transfer_date)
                                            Storage: Memory  Maximum Storage: 17kB
                                            ->  Sort  (cost=0.20..0.21 rows=6 width=8) (actual time=0.009..0.010 rows=6.00 loops=1)
                                                  Sort Key: sales_detail.venue_id, sales_detail.transfer_date
                                                  Sort Method: quicksort  Memory: 25kB
                                                  ->  CTE Scan on sales_detail  (cost=0.00..0.12 rows=6 width=8) (actual time=0.000..0.001 rows=6.00 loops=1)
                                                        Storage: Memory  Maximum Storage: 17kB
Planning:
  Buffers: shared hit=70 read=2
Planning Time: 0.802 ms
Execution Time: 0.261 ms
(67 ♦♦♦♦♦)
```

	Запрос 1 до	Запрос 1 с	Запрос 5 до	Запрос 5 с
Planning Time	1.231 ms	0.671 ms	0.613 ms	0.802 ms
Execution Time	0.335 ms	0.391 ms	0.287 ms	0.261 ms
Scan Types	Seq Scan	Seq Scan	Seq Scan, CTE Scan	Seq Scan, CTE Scan
JOINs	Hash Join, Hash Left Join	Hash Join, Hash Left Join	Hash Join, Hash Left Join	Hash Join, Hash Left Join
```